Class jpeg_decompress_struct

JPEG decompression routine.

Inheritance

↳ System.Object
  ↳ jpeg_common_struct (BitMiracle.LibJpeg.Classic.jpeg_common_struct.html)
    ↳ jpeg_decompress_struct

Inherited Members

jpeg_common_struct.Progress
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_Progress)
jpeg_common_struct.Err
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_Err)
jpeg_common_struct.Version
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_Version)
jpeg_common_struct.Copyright
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_Copyright
)
jpeg_common_struct.CreateSamplesArray(Int32, Int32)
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_CreateSamplesArray_System_Int32_System_Int32_)
jpeg_common_struct.CreateBlocksArray(Int32, Int32)
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_CreateBlocksArray_System_Int32_System_Int32_)
jpeg_common_struct.AllocJpegSamples(Int32, Int32)
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_AllocJpegSamples_System_Int32_System_Int32_)
jpeg_common_struct.jpeg_abort()
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_jpeg_abort)
jpeg_common_struct.jpeg_destroy()
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_jpeg_destroy)
jpeg_common_struct.ERREXIT(J_MESSAGE_CODE)
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_ERREXIT_BitMiracle_LibJpeg_Classic_J_MESSAGE_CODE_)
jpeg_common_struct.ERREXIT(J_MESSAGE_CODE, Object[])
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_ERREXIT_BitMiracle_LibJpeg_Classic_J_MESSAGE_CODE_System_Object___)
jpeg_common_struct.ERREXIT(Int32, Object[])
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_ERREXIT_System_Int32_System_Object___)
jpeg_common_struct.WARNMS(J_MESSAGE_CODE)
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_WARNMS_BitMiracle_LibJpeg_Classic_J_MESSAGE_CODE_)

jpeg_common_struct.WARNMS(J_MESSAGE_CODE, Object[])
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_WARNMS_BitMiracle_LibJpeg_Classic_J_MESSAGE_CODE_System_Object___)
jpeg_common_struct.WARNMS(Int32, Object[])
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_WARNMS_System_Int32_System_Object___)
jpeg_common_struct.TRACEMS(Int32, J_MESSAGE_CODE)
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_TRACEMS_System_Int32_BitMiracle_LibJpeg_Classic_J_MESSAGE_CODE_)
jpeg_common_struct.TRACEMS(Int32, J_MESSAGE_CODE, Object[])
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_TRACEMS_System_Int32_BitMiracle_LibJpeg_Classic_J_MESSAGE_CODE_System_Object___)
jpeg_common_struct.TRACEMS(Int32, Int32, Object[])
(BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_TRACEMS_System_Int32_System_Int32_System_Object___)
System.Object.ToString()
System.Object.Equals(System.Object)
System.Object.Equals(System.Object, System.Object)
System.Object.ReferenceEquals(System.Object, System.Object)
System.Object.GetHashCode()
System.Object.GetType()
System.Object.MemberwiseClone()

Namespace: **BitMiracle.LibJpeg.Classic (BitMiracle.LibJpeg.Classic.html)**

Assembly: **BitMiracle.LibJpeg.NET.dll**

Syntax

```
public class jpeg_decompress_struct : jpeg_common_struct
```

Constructors

jpeg_decompress_struct()

Initializes a new instance of the jpeg_decompress_struct (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html) class.

Declaration

```
public jpeg_decompress_struct()
```

See Also

jpeg_compress_struct (BitMiracle.LibJpeg.Classic.jpeg_compress_struct.html)

jpeg_decompress_struct(jpeg_error_mgr)

Initializes a new instance of the jpeg_decompress_struct (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html) class.

Declaration

```
public jpeg_decompress_struct(jpeg_error_mgr errorManager)
```

Parameters

Type	Name	Description
jpeg_error_mgr (BitMiracle.LibJpeg.Classic.jpeg_error_mgr.html)	*errorManager*	The error manager.

See Also

jpeg_compress_struct (BitMiracle.LibJpeg.Classic.jpeg_compress_struct.html)

Properties

Actual_number_of_colors

The number of colors in the color map.

Declaration

```
public int Actual_number_of_colors { get; set; }
```

Property Value

Type	Description
System.Int32	The number of colors in the color map.

See Also

Colormap (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Colormap)
Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Buffered_image

Gets or sets a value indicating whether to use buffered-image mode.

Declaration

```
public bool Buffered_image { get; set; }
```

Property Value

Type	Description
System.Boolean	`true` if buffered-image mode is turned on; otherwise, `false`.

See Also

Buffered-image mode (../articles/KB/buffered-image-mode.html)

Coef_bits

Gets the current progression status..

Declaration

```
public int[][] Coef_bits { get; }
```

Property Value

Type	Description
System.Int32[][]	`Coef_bits[c][i]` indicates the precision with which component c's DCT coefficient i (in zigzag order) is known. It is `-1` when no data has yet been received, otherwise it is the point transform (shift) value for the most recent scan of the coefficient (thus, 0 at completion of the progression). This is null when reading a non-progressive file.

See Also

Progressive JPEG support (../articles/KB/progressive-jpeg.html)

Colormap

The color map, represented as a 2-D pixel array of Out_color_components (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Out_color_components) rows and Actual_number_of_colors (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Actual_number_of_colors) columns.

Declaration

```
public byte[][] Colormap { get; set; }
```

Property Value

Type	Description
System.Byte[][]	Colormap is set to `null` by jpeg_read_header(Boolean) (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_read_header_System_Boolean_) . The application can supply a color map by setting `Colormap` non-null and setting Actual_number_of_colors (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Actual_number_of_colors) to the map size.

Remarks

Ignored if not quantizing.
Implementation restriction: at present, an externally supplied `Colormap` is only accepted for 3-component output color spaces.

See Also

Actual_number_of_colors (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Actual_number_of_colors)
Quantize_colors (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Quantize_colors)

Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Comp_info

Comp_info[i] describes component that appears i'th in SOF

Declaration

```
public jpeg_component_info[] Comp_info { get; }
```

Property Value

Type	Description
jpeg_component_info (BitMiracle.LibJpeg.Classic.jpeg_component_info.html)[]	The components in SOF.

See Also

jpeg_component_info (BitMiracle.LibJpeg.Classic.jpeg_component_info.html)

Data_precision

Gets the data precision.

Declaration

```
public int Data_precision { get; }
```

Property Value

Type	Description
System.Int32	The data precision.

Dct_method

Gets or sets the algorithm used for the DCT step.

Declaration

```
public J_DCT_METHOD Dct_method { get; set; }
```

Property Value

Type	Description
J_DCT_METHOD (BitMiracle.LibJpeg.Classic.J_DCT_METHOD.html)	The algorithm used for the DCT step.

See Also

Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Density_unit

Gets the resolution information from JFIF marker.

Declaration

```
public DensityUnit Density_unit { get; }
```

Property Value

Type	Description
DensityUnit (BitMiracle.LibJpeg.Classic.DensityUnit.html)	The information from JFIF marker.

See Also

X_density (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_X_density)
Y_density (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Y_density)
Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Desired_number_of_colors

Maximum number of colors to use in generating a library-supplied color map.

Declaration

```
public int Desired_number_of_colors { get; set; }
```

Property Value

Type	Description
System.Int32	Default value: 256.

Remarks

Ignored if Quantize_colors (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Quantize_colors) is `false` .
The actual number of colors is returned in a Actual_number_of_colors (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Actual_number_of_colors).

See Also

Quantize_colors (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Quantize_colors)
Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Dither_mode

Selects color dithering method.

Declaration

```
public J_DITHER_MODE Dither_mode { get; set; }
```

Property Value

Type	Description
J_DITHER_MODE (BitMiracle.LibJpeg.Classic.J_DITHER_MODE.html)	Default value: JDITHER_FS (BitMiracle.LibJpeg.Classic.J_DITHER_MODE.html#BitMiracle_LibJpeg_Classic_J_DITHER_MODE_JDITHER_FS) .

Remarks

Ignored if Quantize_colors (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Quantize_colors) is `false` .
At present, ordered dither is implemented only in the single-pass, standard-colormap case. If you ask for ordered dither when Two_pass_quantize (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Two_pass_quantize) is `true` or when you supply an external color map, you'll get F-S dithering.

See Also

Quantize_colors (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Quantize_colors)
Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Do_block_smoothing

Apply interblock smoothing in early stages of decoding progressive JPEG files.

Declaration

```
public bool Do_block_smoothing { get; set; }
```

Property Value

Type	Description
System.Boolean	If `true` , interblock smoothing is applied in early stages of decoding progressive JPEG files; if `false` , not. Early progression stages look "fuzzy" with smoothing, "blocky" without.

Remarks

Default value: `true`
In any case, block smoothing ceases to be applied after the first few AC coefficients are known to full accuracy, so it is relevant only when using buffered-image mode (../articles/KB/buffered-image-mode.html) for progressive images.

See Also

Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Do_fancy_upsampling

Enable or disable upsampling of chroma components.

Declaration

```
public bool Do_fancy_upsampling { get; set; }
```

Property Value

Type	Description
System.Boolean	If `true`, do careful upsampling of chroma components. If `false`, a faster but sloppier method is used. The visual impact of the sloppier method is often very small.

Remarks

Default value: `true`

See Also

Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Enable_1pass_quant

Enable future use of 1-pass quantizer.

Declaration

```
public bool Enable_1pass_quant { get; set; }
```

Property Value

Type	Description
System.Boolean	Default value: `false`

Remarks

Significant only in buffered-image mode.

See Also

Buffered-image mode (../articles/KB/buffered-image-mode.html)

Enable_2pass_quant

Enable future use of 2-pass quantizer.

Declaration

```
public bool Enable_2pass_quant { get; set; }
```

Property Value

Type	Description
System.Boolean	Default value: `false`

Remarks

Significant only in buffered-image mode.

See Also

Buffered-image mode (../articles/KB/buffered-image-mode.html)

Enable_external_quant

Enable future use of external colormap.

Declaration

```
public bool Enable_external_quant { get; set; }
```

Property Value

Type	Description
System.Boolean	Default value: `false`

Remarks

Significant only in buffered-image mode.

See Also

Buffered-image mode (../articles/KB/buffered-image-mode.html)

Image_height

Gets the height of image, set by jpeg_read_header(Boolean) (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_read_header_System_Boolean_)

Declaration

```
public int Image_height { get; }
```

Property Value

Type	Description
System.Int32	The height of image.

See Also

Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Image_width

Gets the width of image, set by jpeg_read_header(Boolean) (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_read_header_System_Boolean_)

Declaration

```
public int Image_width { get; }
```

Property Value

Type	Description
System.Int32	The width of image.

See Also

Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Input_iMCU_row

Gets the number of iMCU rows completed.

Declaration

```
public int Input_iMCU_row { get; }
```

Property Value

Type	Description
System.Int32	The number of iMCU rows completed.

Remarks

Indicates the progress of the decompressor input side.

Input_scan_number

Gets the number of SOS markers seen so far.

Declaration

```
public int Input_scan_number { get; }
```

Property Value

Type	Description
System.Int32	The number of SOS markers seen so far.

Remarks

Indicates the progress of the decompressor input side.

IsDecompressor

Retrieves `true` because this is a decompressor.

Declaration

```
public override bool IsDecompressor { get; }
```

Property Value

Type	Description
System.Boolean	`true`

Overrides

jpeg_common_struct.IsDecompressor (BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_IsDecompressor)

Jpeg_color_space

Gets or sets the colorspace of JPEG image.

Declaration

```
public J_COLOR_SPACE Jpeg_color_space { get; set; }
```

Property Value

Type	Description
J_COLOR_SPACE (BitMiracle.LibJpeg.Classic.J_COLOR_SPACE.html)	The colorspace of JPEG image.

See Also

Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Marker_list

Gets the list of loaded special markers.

Declaration

```
public ReadOnlyCollection<jpeg_marker_struct> Marker_list { get; }
```

Property Value

Type	Description
System.Collections.ObjectModel.ReadOnlyCollection< jpeg_marker_struct (BitMiracle.LibJpeg.Classic.jpeg_marker_struct.html)>	The list of loaded special markers.

Remarks

All the special markers in the file appear in this list, in order of their occurrence in the file (but omitting any markers of types you didn't ask for)

See Also

Special markers (../articles/KB/special-markers.html)

Max_v_samp_factor

Gets the largest vertical sample factor.

Declaration

```
public int Max_v_samp_factor { get; }
```

Property Value

Type	Description
System.Int32	The largest vertical sample factor.

Num_components

Gets the number of color components in JPEG image.

Declaration

```
public int Num_components { get; }
```

Property Value

Type	Description
System.Int32	The number of color components.

See Also

Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Out_color_components

Gets the number of color components in Out_color_space (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Out_color_space).

Declaration

```
public int Out_color_components { get; }
```

Property Value

Type	Description
System.Int32	The number of color components.

Remarks

Computed by jpeg_start_decompress() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_start_decompress). You can also use jpeg_calc_output_dimensions() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_calc_output_dimensions) to determine this value in advance of calling jpeg_start_decompress() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_start_decompress).

See Also

Out_color_space (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Out_color_space)
Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Out_color_space

Gets or sets the output color space.

Declaration

```
public J_COLOR_SPACE Out_color_space { get; set; }
```

Property Value

Type	Description
J_COLOR_SPACE (BitMiracle.LibJpeg.Classic.J_COLOR_SPACE.html)	The output color space.

See Also

Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Output_components

Gets the number of color components returned.

Declaration

```
public int Output_components { get; }
```

Property Value

Type	Description
System.Int32	When Quantize_colors (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Quantize_colors), `Output_components` is 1, indicating a single color map index per pixel. Otherwise it equals to Out_color_components (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Out_color_components).

Remarks

Computed by jpeg_start_decompress() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_start_decompress). You can also use jpeg_calc_output_dimensions() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_calc_output_dimensions) to determine this value in advance of calling jpeg_start_decompress() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_start_decompress).

See Also

Out_color_space (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Out_color_space)
Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Output_height

Gets the actual height of output image.

Declaration

```
public int Output_height { get; }
```

Property Value

Type	Description
System.Int32	The height of output image.

Remarks

Computed by jpeg_start_decompress() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_start_decompress). You can also use jpeg_calc_output_dimensions() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_calc_output_dimensions) to determine this value in advance of calling jpeg_start_decompress() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_start_decompress).

See Also

Output_width (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Output_width)

Output_iMCU_row

Gets the number of iMCU rows read.

Declaration

```
public int Output_iMCU_row { get; }
```

Property Value

Type	Description
System.Int32	The number of iMCU rows read.

Output_scan_number

Gets the nominal scan number being displayed.

Declaration

```
public int Output_scan_number { get; }
```

Property Value

Type	Description
System.Int32	The nominal scan number being displayed.

Output_scanline

Gets the number of scanlines returned so far.

Declaration

```
public int Output_scanline { get; }
```

Property Value

Type	Description
System.Int32	The output_scanline.

Remarks

Usually you can just use this variable as the loop counter, so that the loop test looks like `while (cinfo.Output_scanline < cinfo.Output_height)`

See Also

Decompression details (../articles/KB/decompression-details.html)

Output_width

Gets the actual width of output image.

Declaration

```
public int Output_width { get; }
```

Property Value

Type	Description
System.Int32	The width of output image.

Remarks

Computed by jpeg_start_decompress() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_start_decompress). You can also use jpeg_calc_output_dimensions() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_calc_output_dimensions) to determine this value in advance of calling jpeg_start_decompress() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_start_decompress).

See Also

Output_height (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Output_height)

Quantize_colors

Colors quantization.

Declaration

```
public bool Quantize_colors { get; set; }
```

Property Value

Type	Description
System.Boolean	If set `true`, colormapped output will be delivered. Default value: `false`, meaning that full-color output will be delivered.

See Also

Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Raw_data_out

Enable or disable raw data output.

Declaration

```
public bool Raw_data_out { get; set; }
```

Property Value

Type	Description
System.Boolean	`true` if raw data output is enabled; otherwise, `false`.

Remarks

Default value: `false`
Set this to true before jpeg_start_decompress() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_start_decompress) if you need to obtain raw data output.

See Also

jpeg_read_raw_data(Byte[][][], Int32) (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_read_raw_data_System_Byte_____System_Int32_)

Rec_outbuf_height

Gets the recommended height of scanline buffer.

Declaration

```
public int Rec_outbuf_height { get; }
```

Property Value

Type	Description
System.Int32	In high-quality modes, `Rec_outbuf_height` is always 1, but some faster, lower-quality modes set it to larger values (typically 2 to 4).

Remarks

Computed by jpeg_start_decompress() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_start_decompress). You can also use jpeg_calc_output_dimensions() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_calc_output_dimensions) to determine this value in advance of calling jpeg_start_decompress() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_start_decompress).

`Rec_outbuf_height` is the recommended minimum height (in scanlines) of the buffer passed to jpeg_read_scanlines(Byte[][], Int32) (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_read_scanlines_System_Byte____System_Int32_). If the buffer is smaller, the library will still work, but time will be wasted due to unnecessary data copying. If you are going to ask for a high-speed processing mode, you may as well go to the trouble of honoring `Rec_outbuf_height` so as to avoid data copying. (An output buffer larger than `Rec_outbuf_height` lines is OK, but won't provide any material speed improvement over that height.)

See Also

Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Scale_denom

Gets or sets the denominator of the fraction of image scaling.

Declaration

```
public int Scale_denom { get; set; }
```

Property Value

Type	Description
System.Int32	Scale the image by the fraction Scale_num (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Scale_num) /Scale_denom. Default is 1/1, or no scaling. Currently, the only supported scaling ratios are 1/1, 1/2, 1/4, and 1/8. (The library design allows for arbitrary scaling ratios but this is not likely to be implemented any time soon.)

Remarks

Smaller scaling ratios permit significantly faster decoding since fewer pixels need to be processed and a simpler J_DCT_METHOD (BitMiracle.LibJpeg.Classic.J_DCT_METHOD.html) can be used.

See Also

Scale_num (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Scale_num)
Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Scale_num

Gets or sets the numerator of the fraction of image scaling.

Declaration

```
public int Scale_num { get; set; }
```

Property Value

Type	Description
System.Int32	Scale the image by the fraction Scale_num/Scale_denom (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Scale_denom). Default is 1/1, or no scaling. Currently, the only supported scaling ratios are 1/1, 1/2, 1/4, and 1/8. (The library design allows for arbitrary scaling ratios but this is not likely to be implemented any time soon.)

Remarks

Smaller scaling ratios permit significantly faster decoding since fewer pixels need to be processed and a simpler J_DCT_METHOD (BitMiracle.LibJpeg.Classic.J_DCT_METHOD.html) can be used.

See Also

Scale_denom (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Scale_denom)
Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Src

Gets or sets the source for decompression.

Declaration

```
public jpeg_source_mgr Src { get; set; }
```

Property Value

Type	Description
jpeg_source_mgr (BitMiracle.LibJpeg.Classic.jpeg_source_mgr.html)	The source for decompression.

Two_pass_quantize

Gets or sets a value indicating whether to use two-pass color quantization.

Declaration

```
public bool Two_pass_quantize { get; set; }
```

Property Value

Type	Description
System.Boolean	If `true` , an extra pass over the image is made to select a custom color map for the image. This usually looks a lot better than the one-size-fits-all colormap that is used otherwise. Ignored when the application supplies its own color map. Default value: `true`

Remarks

Ignored if Quantize_colors (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Quantize_colors) is `false` .

See Also

Quantize_colors (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Quantize_colors)
Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

Unread_marker

Gets the last read and unprocessed JPEG marker.

Declaration

```
public int Unread_marker { get; }
```

Property Value

Type	Description
System.Int32	It is either zero or the code of a JPEG marker that has been read from the data source, but has not yet been processed.

See Also

jpeg_set_marker_processor(Int32, jpeg_decompress_struct.jpeg_marker_parser_method) (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_set_marker_processor_System_Int32_BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_marker_parser_method_)
Special markers (../articles/KB/special-markers.html)

X_density

Gets the horizontal component of pixel ratio.

Declaration

```
public short X_density { get; }
```

Property Value

Type	Description
System.Int16	The horizontal component of pixel ratio.

See Also

Y_density (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Y_density)
Density_unit (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Density_unit)

Y_density

Gets the vertical component of pixel ratio.

Declaration

```
public short Y_density { get; }
```

Property Value

Type	Description
System.Int16	The vertical component of pixel ratio.

See Also

X_density (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_X_density)
Density_unit (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Density_unit)

Methods

jpeg_abort_decompress()

Aborts processing of a JPEG decompression operation.

Declaration

```
public void jpeg_abort_decompress()
```

See Also

jpeg_abort (BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_jpeg_abort)()

jpeg_calc_output_dimensions()

Pre-calculate output image dimensions and related values for current decompression parameters.

Declaration

```
public void jpeg_calc_output_dimensions()
```

Remarks

This is allowed for possible use by application. Hence it mustn't do anything that can't be done twice. Also note that it may be called before the master module is initialized!

jpeg_consume_input()

Consume data in advance of what the decompressor requires.

Declaration

```
public ReadResult jpeg_consume_input()
```

Returns

Type	Description
ReadResult (BitMiracle.LibJpeg.Classic.ReadResult.html)	The result of data consumption.

Remarks

This routine can be called at any time after initializing the JPEG object. It reads some additional data and returns when one of the indicated significant events occurs. If called after the EOI marker is reached, it will immediately return JPEG_REACHED_EOI (BitMiracle.LibJpeg.Classic.ReadResult.html#BitMiracle_LibJpeg_Classic_ReadResult_JPEG_REACHED_EOI) without attempting to read more data.

jpeg_copy_critical_parameters(jpeg_compress_struct)

Initializes the compression object with default parameters, then copy from the source object all parameters needed for lossless transcoding.

Declaration

```
public void jpeg_copy_critical_parameters(jpeg_compress_struct dstinfo)
```

Parameters

Type	Name	Description
jpeg_compress_struct (BitMiracle.LibJpeg.Classic.jpeg_compress_struct.html)	*dstinfo*	Target JPEG compression object.

Remarks

Parameters that can be varied without loss (such as scan script and Huffman optimization) are left in their default states.

jpeg_finish_decompress()

Finish JPEG decompression.

Declaration

```
public bool jpeg_finish_decompress()
```

Returns

Type	Description
System.Boolean	Returns `false` if suspended. The return value need be inspected only if a suspending data source is used.

Remarks

This will normally just verify the file trailer and release temp storage.

See Also

jpeg_start_decompress()
(BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_start_decompress)
Decompression details (../articles/KB/decompression-details.html)

jpeg_finish_output()

Finish up after an output pass in buffered-image mode (../articles/KB/buffered-image-mode.html).

Declaration

```
public bool jpeg_finish_output()
```

Returns

Type	Description
System.Boolean	Returns `false` if suspended. The return value need be inspected only if a suspending data source is used.

See Also

jpeg_start_output(Int32)
(BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_start_output_System_Int32_)
Buffered-image mode (../articles/KB/buffered-image-mode.html)

jpeg_has_multiple_scans()

Is there more than one scan?

Declaration

```
public bool jpeg_has_multiple_scans()
```

Returns

Type	Description
System.Boolean	`true` if image has more than one scan; otherwise, `false`

Remarks

If you are concerned about maximum performance on baseline JPEG files, you should use buffered-image mode (../articles/KB/buffered-image-mode.html) only when the incoming file actually has multiple scans. This can be tested by calling this method.

jpeg_input_complete()

Indicates if we have finished reading the input file.

Declaration

```
public bool jpeg_input_complete()
```

Returns

Type	Description
System.Boolean	`true` if we have finished reading the input file.

See Also

Buffered-image mode (../articles/KB/buffered-image-mode.html)

jpeg_read_coefficients()

Read or write the raw DCT coefficient arrays from a JPEG file (useful for lossless transcoding).

Declaration

```
public jvirt_array<JBLOCK>[] jpeg_read_coefficients()
```

Returns

Type	Description
jvirt_array (BitMiracle.LibJpeg.Classic.jvirt_array-1.html) <JBLOCK (BitMiracle.LibJpeg.Classic.JBLOCK.html)>[]	Returns `null` if suspended. This case need be checked only if a suspending data source is used.

Remarks

jpeg_read_header(Boolean) (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_read_header_System_Boolean_) must be completed before calling this.

The entire image is read into a set of virtual coefficient-block arrays, one per component. The return value is an array of virtual-array descriptors.

An alternative usage is to simply obtain access to the coefficient arrays during a buffered-image mode (../articles/KB/buffered-image-mode.html) decompression operation. This is allowed after any jpeg_finish_output() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_finish_output) call. The arrays can be accessed until jpeg_finish_decompress() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_finish_decompress) is called. Note that any call to the library may reposition the arrays, so don't rely on Access(Int32, Int32) (BitMiracle.LibJpeg.Classic.jvirt_array-1.html#BitMiracle_LibJpeg_Classic_jvirt_array_1_Access_System_Int32_System_Int32_) results to stay valid across library calls.

jpeg_read_header(Boolean)

Decompression startup: this will read the source datastream header markers, up to the beginning of the compressed data proper.

Declaration

```
public ReadResult jpeg_read_header(bool require_image)
```

Parameters

Type	Name	Description
System.Boolean	*require_image*	Read a description of **Return Value**.

Returns

Type	Description
Read Result (BitMiracle.LibJpeg.Classic.ReadResult.html)	If you pass `require_image=true` (normal case), you need not check for a JPEG_HEADER_TABLES_ONLY (BitMiracle.LibJpeg.Classic.ReadResult.html#BitMiracle_LibJpeg_Classic_ReadResult_JPEG_HEADER_TABLES_ONLY) return code; an abbreviated file will cause an error exit. JPEG_SUSPENDED (BitMiracle.LibJpeg.Classic.ReadResult.html#BitMiracle_LibJpeg_Classic_ReadResult_JPEG_SUSPENDED) is only possible if you use a data source module that can give a suspension return. This method will read as far as the first SOS marker (ie, actual start of compressed data), and will save all tables and parameters in the JPEG object. It will also initialize the decompression parameters to default values, and finally return JPEG_HEADER_OK (BitMiracle.LibJpeg.Classic.ReadResult.html#BitMiracle_LibJpeg_Classic_ReadResult_JPEG_HEADER_OK). On return, the application may adjust the decompression parameters and then call jpeg_start_decompress() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_start_decompress). (Or, if the application only wanted to determine the image parameters, the data need not be decompressed. In that case, call jpeg_abort() (BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_jpeg_abort) to release any temporary space.) If an abbreviated (tables only) datastream is presented, the routine will return JPEG_HEADER_TABLES_ONLY (BitMiracle.LibJpeg.Classic.ReadResult.html#BitMiracle_LibJpeg_Classic_ReadResult_JPEG_HEADER_TABLES_ONLY) upon reaching EOI. The application may then re-use the JPEG object to read the abbreviated image datastream(s). It is unnecessary (but OK) to call jpeg_abort() (BitMiracle.LibJpeg.Classic.jpeg_common_struct.html#BitMiracle_LibJpeg_Classic_jpeg_common_struct_jpeg_abort) in this case. The JPEG_SUSPENDED (BitMiracle.LibJpeg.Classic.ReadResult.html#BitMiracle_LibJpeg_Classic_ReadResult_JPEG_SUSPENDED) return code only occurs if the data source module requests suspension of the decompressor. In this case the application should load more source data and then re-call `jpeg_read_header` to resume processing.

Type	Description
	If a non-suspending data source is used and `require_image` is `true`, then the return code need not be inspected since only JPEG_HEADER_OK (BitMiracle.LibJpeg.Classic.ReadResult.html#BitMiracle_LibJpeg_Classic_ReadResult_JPEG_HEADER_OK) is possible.

Remarks

Need only initialize JPEG object and supply a data source before calling.
On return, the image dimensions and other info have been stored in the JPEG object. The application may wish to consult this information before selecting decompression parameters.
This routine is now just a front end to jpeg_consume_input() (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_consume_input), with some extra error checking.

See Also

Decompression details (../articles/KB/decompression-details.html)
Decompression parameter selection (../articles/KB/decompression-parameter-selection.html)

jpeg_read_raw_data(Byte[][][], Int32)

Alternate entry point to read raw data.

Declaration

```
public int jpeg_read_raw_data(byte[][][] data, int max_lines)
```

Parameters

Type	Name	Description
System.Byte[][][]	*data*	The raw data.
System.Int32	*max_lines*	The number of scanlines for reading.

Returns

Type	Description
System.Int32	The number of lines actually read.

Remarks

Replaces jpeg_read_scanlines(Byte[][], Int32) (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_read_scanlines_System_Byte____System_Int32_) when reading raw downsampled data. Processes exactly one iMCU row per call, unless suspended.

jpeg_read_scanlines(Byte[][], Int32)

Read some scanlines of data from the JPEG decompressor.

Declaration

```
public int jpeg_read_scanlines(byte[][] scanlines, int max_lines)
```

Parameters

Type	Name	Description
System.Byte[][]	*scanlines*	Buffer for filling.
System.Int32	*max_lines*	Required number of lines.

Returns

Type	Description
System.Int32	The return value will be the number of lines actually read. This may be less than the number requested in several cases, including bottom of image, data source suspension, and operating modes that emit multiple scanlines at a time.

Remarks

We warn about excess calls to `jpeg_read_scanlines` since this likely signals an application programmer error. However, an oversize buffer `(max_lines > scanlines remaining)` is not an error.

See Also

Decompression details (../articles/KB/decompression-details.html)

jpeg_save_markers(Int32, Int32)

Control saving of COM and APPn markers into Marker_list (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Marker_list).

Declaration

```
public void jpeg_save_markers(int marker_code, int length_limit)
```

Parameters

Type	Name	Description
System.Int32	*marker_code*	The marker type to save (see JPEG_MARKER enumeration). To arrange to save all the special marker types, you need to call this routine 17 times, for COM and APP0-APP15 markers.
System.Int32	*length_limit*	If the incoming marker is longer than `length_limit` data bytes, only `length_limit` bytes will be saved; this parameter allows you to avoid chewing up memory when you only need to see the first few bytes of a potentially large marker. If you want to save all the data, set `length_limit` to 0xFFFF; that is enough since marker lengths are only 16 bits. As a special case, setting `length_limit` to 0 prevents that marker type from being saved at all. (That is the default behavior, in fact.)

See Also

Marker_list
(BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_Marker_list)
Special markers (../articles/KB/special-markers.html)

jpeg_set_marker_processor(Int32, jpeg_decompress_struct.jpeg_marker_parser_method)

Sets processor for special marker.

Declaration

```
public void jpeg_set_marker_processor(int marker_code, jpeg_decompress_struct.jpeg_marker_parser_method routine)
```

Parameters

Type	Name	Description
System.Int32	*marker_code*	The marker code.
jpeg_decompress_struct.jpeg_marker_parser_method (BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.jpeg_marker_parser_method.html)	*routine*	The processor.

Remarks

Allows you to supply your own routine to process COM and/or APPn markers on-the-fly as they are read.

See Also

Special markers (../articles/KB/special-markers.html)

jpeg_start_decompress()

Decompression initialization.

Declaration

```
public bool jpeg_start_decompress()
```

Returns

Type	Description
System.Boolean	Returns `false` if suspended. The return value need be inspected only if a suspending data source is used.

Remarks

jpeg_read_header(Boolean)
(BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_read_header_System_Boolean_) must be completed before calling this.

```
If a multipass operating mode was selected, this will do all but the last pass, and thus may take a great deal of time.
```

See Also

jpeg_finish_decompress()
(BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_finish_decompress)
Decompression details (../articles/KB/decompression-details.html)

jpeg_start_output(Int32)

Initialize for an output pass in buffered-image mode (../articles/KB/buffered-image-mode.html).

Declaration

```
public bool jpeg_start_output(int scan_number)
```

Parameters

Type	Name	Description
System.Int32	*scan_number*	Indicates which scan of the input file is to be displayed; the scans are numbered starting at 1 for this purpose.

Returns

Type	Description
System.Boolean	`true` if done; `false` if suspended

See Also

jpeg_finish_output()
(BitMiracle.LibJpeg.Classic.jpeg_decompress_struct.html#BitMiracle_LibJpeg_Classic_jpeg_decompress_struct_jpeg_finish_output)
Buffered-image mode (../articles/KB/buffered-image-mode.html)

jpeg_stdio_src(Stream)

Sets input stream.

Declaration

```
public void jpeg_stdio_src(Stream infile)
```

Parameters

Type	Name	Description
System.IO.Stream	*infile*	The input stream.

Remarks

The caller must have already opened the stream, and is responsible for closing it after finishing decompression.

See Also

Decompression details (../articles/KB/decompression-details.html)

See Also

jpeg_compress_struct (BitMiracle.LibJpeg.Classic.jpeg_compress_struct.html)